

THE ROLE OF DEMOGRAPHIC FEEDBACK ON THE EVOLUTION OF DEVELOPMENTAL WINDOWS

DATE

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Part I

Main Manuscript

1 Introduction

1.1 Main question with examples

The main question is, given the predictability and severity of food limitations in an organism's environment, over what length of time do individuals invest in developing

higher reproductive quality at an opportunity cost to investing in growth to exploit periods of abundant food?

Developmental windows are periods in development where sensitivity to cues is heightened.

Walasek et al. (2021), Walasek et al. (2022) find that the timing of when sensitive periods emerge depends on the reliability of cues increases across ontogeny, and their results suggest that both information and phenotypic states are modelled, with incremental development in fluctuating environments favouring residual development across ontogeny.

Further, Walasek et al (2024) review current theory and empirical literature and suggest future models could better integrate with empirical research by incorporating phenomena found in empirical systems into our models, and design more models that can be tailored to the life histories of specific species. They encourage for instance that models that incorporate the life history trade-offs we see in the development of alternative phenotypes in bulb mites (between scramblers and fighters) (Smallegange, 2023; Smallegange et al., 2019), and it happens that authors interested in these taxa are also interested in bridging the gap in research between demographic modelling and agency with respect to development shaping demography (Smallegange).

1.2 Developmental windows and sensitive periods

1.3 Demography, agency and life-history

Feedback between plastic responses and population dynamics has been explored for fixed reaction norms by Brass et al. (2021) who showed that _____. However, it is unclear

how these eco-devo dynamics will play out over evolutionary time.

1.4 Hypothesis

1.5 Methods

2 The Model Outline

We model a large population of asexual individuals, distributed in a metapopulation across habitat patches of equal size and quality, but each with a population density state with transition probabilities determined by the local population's growth rate. Each individual has three state dimensions—size (s), patch information (π), and reproductive quality (m)—upon which their developmental strategy is dependent, $x(s, \pi, m)$, which determines the fraction of ingested energy allocated to developing greater reproductive quality over growth to increase their size state (and so ultimately the local population growth rate). Size determines whether or not an individual is mature given they can only reproduce when they reach their maximum size S . An individual's patch information state gives them the subjective probabilities regarding each possible population density state of their habitat patch. Finally, an individual's reproductive quality state m determines their rate of reproduction at maturity.

We assume mutations are rare and of small effect such that effectively all individuals across the metapopulation have the same state-dependent developmental strategy (henceforth referred to as the resident strategy). Given the resident strategy we then do a forward projection using a Markov chain until demographic stability is reached at the metapopulation level (i.e. among patches). Given the probability distribution of

population density states of patches, and assuming individuals have complete information as to this probability distribution and of the transition probabilities between patch pop. density states, we find the best response strategy for an invading individual (i.e. the developmental strategy that maximises their expected fitness).

2.1 The Life Cycle and Demographic Feedback

We model the life cycle of both individuals and of the patches they reside in (see figure 1).

1. Newborns disperse into random patches, with accurate prior information as to the probability distribution over patch population density states, and causing transitions in patch population density states.
2. Individuals forage, with food availability being a random variable, normally distributed, with mean determined by the population density.
3. Individuals update their patch information state given informativeness of food availability.
4. Immature individuals allocate a proportion of their ingested energy towards increasing their future reproductive quality given their developmental strategy, the rest of the energy goes towards maturation (increasing their size). Mature individuals do not grow, investing all ingested energy towards current reproduction.
5. Individuals die given a baseline probability of death from exogenous factors and given perturbations in patch quality (e.g. disease outbreaks) influenced by the patch population density state, and in turn causing transitions of patch states to lower population densities.

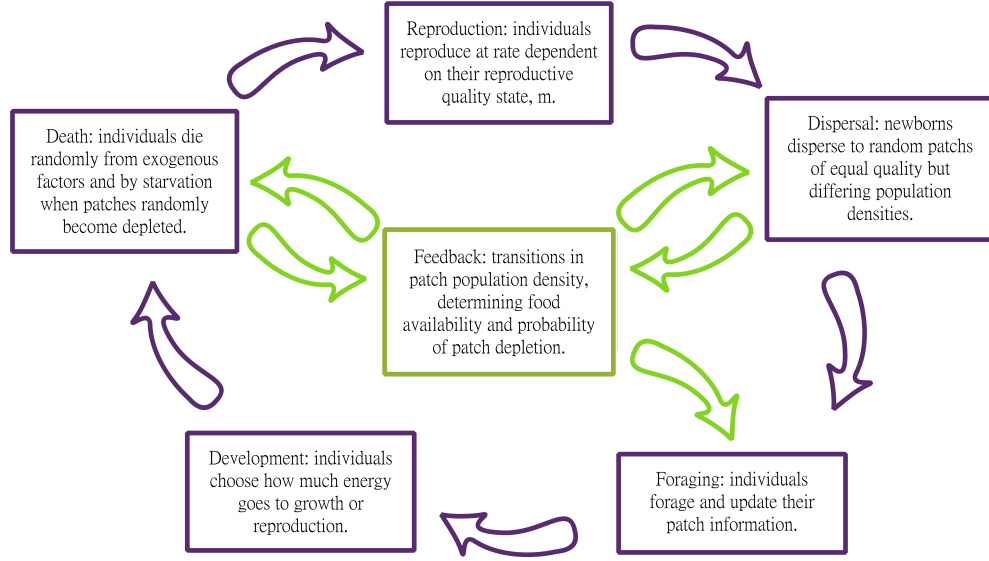


Figure 1: *The model structure, including the life cycle of individuals in purple and the feedback loops acting on patches in green.*

6. Individuals that are of maximum size are mature and reproduce at rate determined by their reproductive quality state.

2.2 State dimensions

- Patch population density state, $i \in \{0, 1, 2\}$, corresponding to low, medium, and high population densities respectively. We assume that i determines the probability of patch quality decline and determining the availability of food.

- Size state, $s \in \{1, 2, \dots, S\}$, where S is their size at maturity.
- Patch belief state, $\pi = \{\pi_0, \pi_1, \pi_2\}$, is an individual's implicit and subjective probability distribution over patch population density states, where $\pi_i = P(i)$, $\pi_i \in \{0/L, 1/L, \dots, L/L\}$, and $\pi_0 + \pi_1 + \pi_2 = 1$, where L determines the resolution of the belief state. Thus, note that π is best represented as co-ordinates within a discretised probability 2-simplex (i.e. an equilateral triangle where each vertex represents each possible state of certainty), rather than representing π with a 3D cube (which would include impossible probability distributions and so waste state space).
- Reproductive quality state, $m \in \{0, 1/M, 2/M, \dots, M/M\}$, corresponding to the quality of an individual's physiology and so the mass of their offspring. Further, any investment in quality above the quality threshold m_{switch} , results in an individual developing an alternative fighter phenotype, giving additional reproductive benefit $r_{fighter}$.

Part II

Appendix

3 Finding Demographic Stability

Here we describe pseudocode for calculating the demographically stable probability distribution of population densities across patches and of the transition probabilities between patch density states. Patches have a population size state $i \in \{0, 1, \dots, I\}$ where

I is the fineness of the grid of possible population sizes. The actual population size for any given state q_i is defined as

$$q_i = n_{max} \cdot \frac{i}{I} \quad (1)$$

where n_{max} is the maximum population size of any patch.

3.1 Reproduction

Adults, individuals of size state $s = S$, reproduce asexually at a rate dependent on their reproductive quality state. We here define $R(s, m)$, the expected reproductive rate of an individual of state $\{s, m\}$ in a patch of density q_i , as

$$R(s, m, i) = \begin{cases} \text{if } s \geq S \text{ and } m \geq m_{switch} : & \sum_g p(g|q_i) \cdot g \cdot (r_b + m \cdot r_m + r_{fighter}) \\ \text{if } s \geq S \text{ and } m < m_{switch} & \sum_g p(g|q_i) \cdot g \cdot (r_b + m \cdot r_m) \\ \text{else :} & 0 \end{cases} \quad (2)$$

where S is their size at maturity, m_{switch} is the threshold investment in quality above which the fighter polyphenism is expressed, r_b is the baseline reproductive rate, r_m is the reproductive benefit of greater somatic quality, and $r_{fighter}$ is reproductive benefit of expressing the fighter polymorphism allowing them to extract auxilliary resources for reproduction. To make solving the model tractable we must assume the fighter phenotype does not influence competition for the shared limited resource represented by g .

3.2 Dispersal

3.2.1 Migrant Physiology

Ideally, individuals would be born with precise and accurate prior information regarding the probability distribution over the population density states of patches. Individuals would then update their information state given cues for the population density of their patch which come from the availability of food, see section 1.2. Individuals are born at size state 0 and reproductive quality state 0, and a belief state over population densities, $\forall i$ of

$$\pi_i \approx h(q_i) \quad (3)$$

However, to form a discrete grid of states that we can solve a Markov decision process for we need interpolate each π_i onto a grid of possible discrete probability values, $\pi_i \in \{0/L, 1/L, 2/L, \dots, L/L\}$, where L is an integer determining the finess of the grid. then needs to be interpolated onto a discrete grid of possible values of π_i which (because it represent a probability distribution) lies on a unit simplex (i.e. all values are positive and they add up to 1), requiring a grid of barycentric co-ordinates. To this end, the first step is to decompose each π'_i into it's integer and fractional parts, with the integer part defined by

$$n_i = \text{floor}(h(q_i) \cdot L),$$

and subsequently the fractional part defined as

$$f_i = h(q_i) \cdot L - n_i.$$

Note that just as

$$\sum_i h(q_i) = 1$$

so does

$$L = \sum_i h(q_i) \cdot L,$$

and so

$$L = \sum_i n_i + f_i$$

and therefore the sum of the fractional parts must equal an integer because L is an integer. Therefore, assuming $\pi = \{\pi_0, \pi_1, \pi_2\}$, the fractional parts must add up to 0, 1 or 2. And so for all states the proportion of dispersors that are of state $\{s, \pi, m\}$ we first set to zero, $\tilde{\rho}(s, \pi, m) = 0$. Then we update given these fractional parts. If the fractional parts add up to 0, all offspring have initial state

$$\tilde{\rho}(s = 1, \pi = \{n_1, n_2, n_3\}, m) = 1$$

Else, if the fractional parts add up to 1 we know exactly one dimension of their prior belief state has to be rounded up, with weightings given to their respective fractional parts and so

$$\tilde{\rho}(s = 1, \pi = \{n_0 + 1, n_1, n_2\}, m = 0) = f_0$$

$$\tilde{\rho}(s = 1, \pi = \{n_0, n_1 + 1, n_2\}, m = 0) = f_1,$$

$$\tilde{\rho}(s = 1, \pi = \{n_0, n_1, n_2 + 1\}, m = 0) = f_2.$$

Else and finally, if the fractional parts add up to 2, exactly 2 dimensions have to be rounded up

$$\tilde{\rho}(s = 1, \pi = \{n_0, n_1 + 1, n_2 + 1\}, m = 0) = 1 - f_0,$$

$$\tilde{\rho}(s = 1, \pi = \{n_0 + 1, n_1, n_2 + 1\}, m = 0) = 1 - f_1,$$

$$\tilde{\rho}(s = 1, \pi = \{n_0 + 1, n_1 + 1, n_2\}, m = 0) = 1 - f_2.$$

3.2.2 Patch Densities transition probabilities

Assuming a dispersal rate of μ and a fixed number of patches N , the number of individuals arriving in each patch of population density q_i is

$$\alpha(q_i) = (1 - \mu) \cdot q_i \cdot r(q_i) + \frac{\mu \cdot N \cdot \sum_{j=0}^I h(q_j) \cdot q_j \cdot r(q_j)}{N} \quad (4)$$

where $r(q)$ is the expected reproductive rate of an individual in a patch of population density q_i and where $h(q_i)$ is the proportion of patches that are of density q_i . Note that N in the numerator of the fraction is there to scale the number of migrants given the number of patches producing them, but that N gets cancelled out as these migrants are dispersed across these patches (see the denominator), telling us that the number of immigrants to a patch is not dependent on the number of patches; only the frequency distribution over patch states matters. The expected reproductive rate of an individual in a patch of population density q_i is defined as

$$r(q_i) = \sum_s^S \sum_\pi^\Pi \sum_m^M [\rho_i(s, \pi, m) \cdot R(s, m, i)] \quad (5)$$

where $\rho_i(s, \pi, m)$ is the probability that an individual will be of state $\{s, \pi, m\}$ given that they are in a patch of population density q_i , and $R(s, m, i)$ is the reproductive rate of an individual of state $\{s, m\}$ in a patch of density q_i (see section 1.4).

Following arrivals, the new population density of patches of prior density q_i become

$$q'_i = \min(q_I, q_i + \alpha(q_i)) \quad (6)$$

We must then interpolate the posterior population density q'_i back on to the grid of density states, with the nearest positions being

$$d_1(i) = \text{floor}\left(\frac{q'_i}{n_{\max}} \cdot I\right) \quad (7)$$

and

$$d_2(i) = d_1(i) + 1. \quad (8)$$

Therefore patches of population density q_i transition to patches of density state $q_{d_2(i)}$ with probability

$$p_2(i) = \left(\frac{q'_i}{n_{max}} \cdot I \right) - d_1(i), \quad (9)$$

and transition to density state $q_{d_1(i)}$ with probability

$$p_1(i) = 1 - p_2(i). \quad (10)$$

This then returns the transition probabilities between patch density states given dispersal $p_{births}(q_{d_1(i)}|q_i)$, which at demographic stability will be used for finding the best response strategy. For all q_i $p_{births}(q_{d_1(i)}|q_i) = 0$ and then

- Add $p_1(i)$ to $p_{births}(q_{d_1(i)}|q_i)$
- Add $p_2(i)$ to $p_{births}(q_{d_2(i)}|q_i)$.

3.2.3 Projection

The posterior frequency distribution over patch population density states is then defined as

$$h'(q_i) = \sum_j h(q_j) \cdot p_{births}(q_i|q_j) \quad (11)$$

Regarding the distribution of physiological states within patches, individuals in any patch state q_i will have a physiological state frequency distribution defined given the transition probabilities between patches and weighted by the change in population size for these transitions. Thus, $\forall i \forall \{s, \pi, m\}$

$$\rho'_i(s, \pi, m) = \sum_j [h(q_j) \cdot p_{births}(q_i|q_j) \cdot \left(\frac{j}{i} \cdot \rho_j(s, \pi, m) + \frac{i-j}{i} \cdot \tilde{\rho}(s, \pi, m) \right)] \quad (12)$$

where if patch were to transition from a density of j to a density of i , $(\frac{j}{i} \cdot \rho_i(s, \pi, m) + \frac{i-j}{i} \cdot \tilde{\rho}(s, \pi, m))$ is the proportion of individuals who are of state $\{s, \pi, m\}$ within that patch. This does however indicate that different patches will have different physiological distributions amongst their populations depending on the historical density states of that patch which would violate our assumptions of this being a Markov process, so (to maintain tractability) we assume that individuals have equal access to food regardless of physiological state.

3.3 Foraging

Individuals observe cues for population density in the form of food availability. Food availability is a random variable influenced by the population density, all patches produce the same amount of food which must be shared, so for any given patch of population density q_i individuals receive a food mass of

$$g \sim \text{Normal}(\frac{\bar{g}}{q_i}, \sigma_g) \quad (13)$$

where parameter $\bar{g}$ is the mean food mass production of any patch and parameter σ_g is the food mass standard deviation. However, as we are working with discrete mathematics, we must approximate this normal distribution using a pseudo-normal probability distribution over a grid of possible food mass values $\mathbf{G} = \{0, 1, \dots, G_{\max}\}$, with probabilities for each realised food mass being

$$p(g|q_i) = \frac{1}{K} e^{-\frac{(g - \frac{\bar{g}}{q_i})^2}{2\sigma_g^2}} \quad \text{for } g \in \mathbf{G}, \quad (14)$$

where N is a suitable integer that determines the fineness of the grid of values, and K is

$$K = \sum_{-G}^G e^{-\frac{(g - \frac{\bar{g}}{q_i})^2}{2\sigma_g^2}}. \quad (15)$$

3.3.1 Observation

Given the food mass g , individuals observe the resulting nutrient data and update their population density information states (i.e. their belief states). Assuming Bayesian optimal updating, their posterior belief is defined $\forall i$ as

$$\pi'_i = \frac{\pi_i \cdot p(g|q_i)}{\sum_j \pi_j \cdot p(g|q_j)}. \quad (16)$$

The posterior belief state π'_i then needs to be interpolated onto a discrete grid of possible values of π_i which (because it represent a probability distribution) lies on a unit simplex (i.e. all values are positive and they add up to 1), requiring a grid of barycentric coordinates. To this end, the first step is to decompose each π'_i into it's integer and fractional parts, with the integer part defined by

$$n_i = \text{floor}(\pi'_i \cdot L),$$

and subsequently the fractional part defined as

$$f_i = \pi'_i \cdot L - n_i.$$

Note that just as

$$\sum_i \pi'_i = 1$$

so does

$$L = \sum_i \pi'_i \cdot L,$$

and so

$$L = \sum_i n_i + f_i$$

and therefore the sum of the fractional parts must equal an integer because L is an integer. Therefore, assuming $\pi = \{\pi_0, \pi_1, \pi_2\}$, the fractional parts must add up to 0, 1 or 2. If they add up to 0

$$p(\pi' = \{n_1, n_2, n_3\} | \pi, g) = 1$$

If the fractional parts add up to 1 we know exactly one dimension of the posterior belief state has to be rounded up, with weightings given to their respective fractional parts and so

$$p(\pi' = \{n_0 + 1, n_1, n_2\} | \pi, g) = f_0$$

$$p(\pi' = \{n_0, n_1 + 1, n_2\} | \pi, g) = f_1,$$

$$p(\pi' = \{n_0, n_1, n_2 + 1\} | \pi, g) = f_2.$$

Else if the fractional parts add up to 2, exactly 2 dimensions have to be rounded up

$$p(\pi' = \{n_0, n_1 + 1, n_2 + 1\} | \pi, g) = 1 - f_0,$$

$$p(\pi' = \{n_0 + 1, n_1, n_2 + 1\} | \pi, g) = 1 - f_1,$$

$$p(\pi' = \{n_0 + 1, n_1 + 1, n_2\} | \pi, g) = 1 - f_2.$$

Finally, for all other possible posterior co-ordinates on the belief grid, the transition probability is

$$p(\pi' | \pi, g) = 0.$$

3.3.2 Development

Each individual, choose a developmental action given their state $\{s, \pi, m\}$, given the resident developmental strategy for that state $x(s, \pi', m) \in (0, G)$. Their developmental action determines the threshold mass above which energy foraged goes to the development of phenotypes of greater reproductive quality (e.g. fighter morphology),

generating a transition in their size state s and their reproductive quality state m . The resulting transition probability, $p(s', \pi', m' | s, \pi, m)$, between state $\{s', \pi', m'\}$ and state $\{s, \pi, m\}$ given the resident developmental strategy $x(s, \pi, m)$, is found by doing the following:

1. For all $\{s, \pi, m, g\}$

$$p(s', m' | s, \pi', m, g) = 0.$$

2. For all g , add 1 to $p(s + \Delta s, m + \Delta m | s, \pi', m, g)$, where

$$\Delta m = \max(g - x(s, \pi', m), 0) \quad (17)$$

is the change in quality state of individuals after ingesting a food mass of g , and where

$$\Delta s = \max(g - \Delta m - y(s, m), 0), \quad (18)$$

is the change in size state of individuals after ingesting a food mass of g , given that all energy allocated to growth below the fixed and state-dependent maintenance cost threshold $y(s, m)$, below which all food mass is invested in maintenance.

3.3.3 Maintenance

Of the portion of energy allocated to quality, all energy below a size-dependent investment threshold goes to maintenance costs. Therefore, if investment in quality falls below this threshold there is a cost to survival

$$P(\text{survival} | g, \Delta m, \Delta s, s, m) = \min(1, \frac{g - \Delta m - \Delta s}{y(s, m)}) \quad (19)$$

such that if the individual has no energy to allocate to maintenance they die, where $y(s)$ is the size dependent maintenance cost

$$y(s, m) = \frac{s^3 \cdot m}{S^3 \cdot M} \cdot g_{\max} \quad (20)$$

where we assume maintenance costs scale by the volume of the individual and by the quality of soma (given their quality state m), and where g_{max} is the maximum maintenance investment.

3.3.4 Updating the States Distributions

Given the baseline rate of death λ and deaths caused by lack of investment in maintenance, the posterior density of each state for patches of each state i is

$$\hat{\rho}'_i(s', \pi', m') = \sum_g \sum_s \sum_{\pi} \sum_m p(s', m' | s, \pi', m, g) \cdot P(survival | s, \pi', m, g) \cdot p(\pi' | \pi, g) \cdot p(g | q_i) \cdot \rho'_i(s, \pi, m) \quad (21)$$

Where, as a reminder, $\rho'_i(s, \pi, m)$ is the probability distribution over states after dispersal within patches of population size q_i , $p(g | q_i)$ is the probability of a food mass of g in patches of size q_i , $p(\pi' | \pi, g)$ is the probability of transitioning to posterior information state π' given this food mass, $P(survival | s, \pi', m, g)$ is the probability of survival of maintenance costs given their developmental strategy dependent on posterior information state π' , and $p(s', m' | s, \pi', m, g)$ is the probability of transitioning to size state s' and quality state m' given the outcomes of development and given that they have survived maintenance costs.

Given the baseline rate of death λ and deaths caused by lack of investment in maintenance, patches can transition to lower population density states. Therefore, given the posterior densities of individuals within patches, the posteior of population size of a patch of density state q_i is

$$u = q_i \cdot (1 - \lambda) \cdot \sum_s \sum_{\pi} \sum_m \hat{\rho}'_i(s', \pi', m')$$

The posterior distribution of patch states is therefore

$$h^\lambda(q_i) = \sum_j^I h(q_j) \cdot p_{\text{fail}}(q_i|q_j) \quad (22)$$

where $p_{\text{fail}}(q_i|q_j)$ is the transition probability of a patch from state j to i given deaths from maintenance failures, defined by interpolating u so that where

$$s_1(j) = \text{floor}\left(\frac{u}{n_{\text{max}}} \cdot I\right) \quad (23)$$

$$s_2(j) = s_1(j) + 1 \quad (24)$$

$$p_2(j) = \left(\frac{u}{n_{\text{max}}} \cdot I\right) - s_1 \quad (25)$$

$$p_1(j) = 1 - p_2(j) \quad (26)$$

and so run the following algorithm for each q_j :

1.

$$\forall q_i, p_{\text{fail}}(q_i|q_j) = 0. \quad (27)$$

2.

$$p_{\text{fail}}(s_1(q_j)|q_j) = h'(q_j) \cdot p_1(j) + p_{\text{fail}}(s_1(q_j)|q_j). \quad (28)$$

3.

$$p_{\text{fail}}(s_2(q_j)|q_j) = h'(q_j) \cdot p_2(j) + p_{\text{fail}}(s_2(q_j)|q_j). \quad (29)$$

The posterior probability distribution over individual states for patches of each population density state is therefore

$$\rho_i''(s', \pi', m') = \sum_{q_j} h(q_j) \cdot p_{\text{fail}}(q_i|q_j) \cdot \frac{\hat{\rho}'_j(s', \pi', m')}{\sum_s \sum_{\pi} \sum_m \hat{\rho}'_j(s', \pi', m')} \quad (30)$$

3.4 Patch quality perturbations

Every patch undergoes momentary perturbations in quality, influenced by patch population density (e.g. the local extinction of a prey species). These perturbations can afford an increase or decrease in population size. For simplicity, we assume patches rapidly return to normal quality after perturbations and we assume that the probability of death is equal for all individuals so there is no change in the distribution of individual states within patches. We assume these perturbations have a normal distribution, although once again we must approximate this normal distribution to interpolate transition probabilities on to the grid of possible density states, where corresponding to the range of possible population density transitions we define the set $T = \{-I, -I + 1, \dots, I - 1, I\}$ (where I is the size of the population density state grid), with a pseudo-normal distribution over this set defining the probability of any given possible change in density $\tau \in T$, defined as

$$p(\tau) = \frac{1}{K_\gamma} e^{-\frac{(\tau/I)^2}{2\sigma_\gamma^2}} \quad \text{for } \tau \in T, \quad (31)$$

where σ_γ^2 is the standard deviation parameter for perturbations and K_γ is

$$K_\gamma = \sum_{t=-I}^I e^{-\frac{(t/I)^2}{2\sigma_\gamma^2}}. \quad (32)$$

The shift in probability generated by demographic feedback is defined as

$$c = \text{round}\left(\frac{q_i}{q_I} \cdot \gamma\right) \quad (33)$$

where $\gamma \in (0, 1)$ is the maximum risk of mass mortality events associated with population density. We can then define $p_{\text{patch}}(q_i|q_j)$, the probability of a patch of density j to a

density of i as a result of patch quality perturbations, as

$$p_{patch}(q_i|q_j) = \begin{cases} \text{if } i = 0 & \sum_{\tau=\min(I,i+c-j)}^{\tau=-I} p(\tau) \\ \text{if } 0 < i < I & p(\tau = \min(I,i+c-j)) \\ \text{if } i = I & \sum_{\tau=\min(I,i+c-j)}^{\tau=I} p(\tau) \end{cases} \quad (34)$$

Given these perturbations, the proportion of patches that are of population density q_i for all i in the next time step is

$$h''(q_i) = \sum_j^I p_{patch}(q_i|q_j) \cdot h'(q_j) \quad (35)$$

3.5 Bayesian updating between growth cycles

Individuals also know the transition probabilities between patch population density states at demographic stability in their evolutionary history, $\tilde{p}(q_i|q_j)$, resulting from these growth and death processes, and so individuals that survive to the next growth cycle will update their information states accordingly for all i

$$\pi'_i = \sum_j \pi_j \cdot \tilde{p}(q_i|q_j). \quad (36)$$

For the first instance, when we have not yet found demographic stability, we will assume that their priors are that $\forall i \forall j, \tilde{p}(q_i|q_j) = 0$, as this should not have a strong influence on the long term adaptive dynamics assuming that the dynamics are ordered rather than chaotic.

3.6 Demographic Stability

Assuming that with successive generations n that the distribution of patches over states converges on stable values, i.e. $\forall q_i$

$$h^*(q_i) = \lim_{n \rightarrow \infty} [h^n(q_i)] \quad (37)$$

further, we can approximate $h^*(q_i)$ by iterating over generations until

$$tol > \sum_i |h(q_i) - h''(q_i)| \quad (38)$$

where the tolerance threshold, tol , is very small, and $h''(q_i)$ is the proportion of patches that are of population density state q_i after dispersal and deaths. Once demographic stability is reached we assume $h^*(q_i) = h(q_i)$ and proceed with invasion analysis.

4 Finding Evolutionary Stability

Having found the demographically stable state of the metapopulation, given the resident strategy, we can then use the Bellman equation to find the best response strategy $b(x(s, \pi, m))$ for an invading individual. We then assume the best response strategy becomes the new resident strategy, and recalculate the demographically stable state of the metapopulation. The evolutionarily stable strategy (ESS) is a resident strategy for which invading individuals can do no better than to follow the resident strategy and so the ESS is defined as

$$\bar{x}(s, \pi, m) = b(x(s, \pi, m)) = x(s, \pi, m). \quad (39)$$

Assuming there is a singular ESSs, we can iterate over invasions till the best response strategy converges on the ESS, and the starting point of the population will not matter.

By sampling from multiple starting points we can test this assumption, and if necessary plot out the adaptive dynamics with phase portraits where there are multiple ESSs or oscillatory dynamics.

To find the best response we define a Bellman fitness equation for individuals in the model where $V_{n=0}(s, \pi, m) = 0$ for all state combinations (as the individual is dead), and iterate backwards to the previous time step $n + 1$

$$V_{n+1}(s, \pi, m) = R(s, m) + \max_{x(s, \pi, m) = \kappa} \left[\sum_{s_n} \sum_{\pi_n} \sum_{m_n} p(\text{survival} | \pi_n) \cdot p(s_n, \pi_n, m_n, \text{survives} | s, \pi, m, \kappa) \cdot V_n(s_n, \pi_n, m_n) \right] \quad (40)$$

where $R(s, m)$ is the rate of reproduction for an individual of size s and quality m (see equation 1), $p(\text{survives} | s, \pi, m)$ is the probability the individual survives between timesteps given maintenance costs, and the baseline probability of death, and $p(s_n, \pi_n, m_n | s, \pi, m, \kappa)$ is the state transition probability to the posterior state $\{s_n, \pi_n, m_n\}$ of individuals of prior state $\{s, \pi, m\}$.

4.1 Dispersal

Individuals are born with accurate prior information regarding the propbability distribution over the population density states of patches

$$\pi = \{\pi_0 = p(q_0) = h(q_0), \pi_1 = p(q_1) = h(q_1), \dots, \pi_I = p(q_I) = h(q_I)\} \quad (41)$$

Individuals then update their information state given cues for the population density of their patch which come from the availability of food. Individuals are born at size state 1 and reproductive quality state 0.

Individuals also know the transition probabilities between patch population density states at demographic stability resulting from dispersal, and so individuals will update

their information state accordingly for all i

$$\pi_i = \sum_j \pi_j \cdot \tilde{p}_{births}(q_i|q_j) \quad (42)$$

where $\tilde{p}_{births}(q_i|q_j)$ is the transition probability at demographic stability of a patch of population density j to a density of i , given dispersal.

4.2 Foraging

Individuals observe cues for population density in the form of food availability. Food availability is a random variable influenced by the population density, all patches produce the same amount of food which must be shared, so for any given patch of population density q_i individuals receive a food mass of

$$g \sim \text{Normal}\left(\frac{\bar{g}}{q_i}, \sigma_g\right) \quad (43)$$

where parameter $\bar{g}$ is the mean food mass production of any patch and parameter σ_g is the food mass standard deviation. However, as we are working with discrete mathematics, we must approximate this normal distribution using a pseudo-normal probability distribution over a grid possible food mass values $G \in \{g_0, g_1, \dots, g_G\}$, with probabilities for each realised food mass being

$$p(g|\pi = p(q_i)) = \frac{1}{K} e^{-\frac{(g - \frac{\bar{g}}{q_i})^2}{2\sigma_g^2}} \quad \text{for } g \in G, \quad (44)$$

where N is a suitable integer that determines the fineness of the grid of values, and K is

$$K = \sum_{-G}^G e^{-\frac{(g - \frac{\bar{g}}{q_i})^2}{2\sigma_g^2}}. \quad (45)$$

4.2.1 Observations

Given the food mass g , individuals observe the resulting nutrient data and update their population density information states. Here we find the transition probability of individuals given these observations

$$p(\pi'|\pi, g) = p(p'(q_j)|p(q_i), g) = \frac{p(q_i) \cdot p(g|p(q_i))}{\sum_k^I p(q_k) \cdot p(g_k|p(q_k))} \quad (46)$$

4.2.2 Development

As observations of patch density and an individual's development are linked by the same random variable, food mass g , we define the state transition probability given individual decision making as

$$a(s', \pi', m', \text{survives}|s, \pi, m, \kappa) = \sum_g p(g|\pi) \cdot p(\pi'|\pi, g) \cdot p(s', m', \text{survives}|\kappa, s, m, \pi', g) \quad (47)$$

where $p(g|\pi)$ is the probability of food mass g given the information state π

$$p(g|\pi) = \sum_i^I \pi_i \cdot p(g|q_i),$$

and where $p(\pi'|\pi, g)$ is the probability of posterior information state π' after an individual of prior information state π observes a food mass of g , and $p(s', m', \text{survives}|\kappa, s, m, \pi', g)$ is the probability of posterior size s' and quality m' given the individual's developmental policy dependent on their current size, quality, and information states, given the food mass they received, and given their probability of survival given maintenance costs, all defined once by doing the following:

1. For all $\{s, \pi, m, g, \kappa\}$

$$p(s', m'|s, \pi, m, g, \kappa) = 0.$$

2. For all i and all g , define the transitions in state predicted by their developmental strategy κ , where

$$\Delta m = \max(g - \kappa, 0) \quad (48)$$

is the change in quality state of individuals after ingesting a food mass of g , and where

$$\Delta s = \max(g - \Delta m - y(s, m), 0), \quad (49)$$

is the change in size state of individuals after ingesting a food mass of g , given that all energy allocated to growth below the fixed and state-dependent maintenance cost threshold $y(s, m)$, below which all food mass is invested in maintenance.

3. Then add $p(\text{survival}|g, \Delta m, \Delta s, s, m)$ to $p(s + \Delta s, m + \Delta m, \text{survives}|s, \pi, m, g, \kappa)$, where depending on the food mass g an individual encounters and the fraction of that mass that they invest in maintenance, there is a probability of death from lack of maintenance of

$$P(\text{survival}|g, \Delta m, \Delta s, s, m) = \min(1, \frac{g - \Delta m - \Delta s}{y(s, m)}) \quad (50)$$

where $y(s, m)$ is the size and quality dependent maintenance cost threshold to avoid death completely, defined as shown previously:

$$y(s, m) = \frac{s^3 \cdot m}{S^3 \cdot M} \cdot g_{\max} \quad (51)$$

So then, accounting for both the demographically stable transition probabilities between patch density states, and given individual decision-making, the probability of an individual transitioning from state $\{s, \pi, m\}$ to state $\{s_n, \pi_n, m_n\}$ is

$$p(s_n, \pi_n, m_n, \text{survives}|s, \pi, m, \kappa) = \sum_{\pi'} p(\pi_n|\pi', p_{\text{deaths}}(q_k|q_i)) \cdot \sum_{\tilde{\pi}} a(s_n, \pi', m_n, \text{survives}|s, \tilde{\pi}, m, \kappa) \cdot p(\tilde{\pi}|\pi, p) \quad (52)$$

where at demographic stability we found transition probabilities for patch density states after deaths and births, $p_{deaths}(q_j|q_k)$ and $p_{births}(q_k|q_i)$ respectively and we assume these transition probabilities are known to the invading individual, generating predictable transitions in their belief state.

4.3 Depletion

Individuals have a probability of death from patch depletion and a probability of death λ from exogenous factors (e.g. old age). Given an individual's information state after foraging $\boldsymbol{\pi}' = \{\pi_{j=1}, \pi_{j=2}, \dots, \pi_{j=I}\}$ and given the probability of patch density transition at demographic stability, the probability that an individual dies due to patch depletion is

$$P(survival|\boldsymbol{\pi}') = (1 - \lambda) \cdot \sum_j \sum_k \pi_j \cdot (1 - p_{deaths}(q_k|q_j) \cdot \frac{j-k}{j}) \quad (53)$$